

Sarthak Avhad

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Professional Summary

M.Sc. Computer Science student specializing in Blockchain Technology at MIT World Peace University, with hands-on project experience spanning blockchain infrastructure (**Solidity, Ethereum, zk-SNARKs, Solana, DeFi**), full-stack web development (MERN stack), and **DevOps/CI-CD** workflows. Comfortable building AI-integrated applications end-to-end, from smart contract and backend logic to automated testing and deployment pipelines, with a strong interest in decentralized systems and privacy-preserving protocols.

Projects

TestGen-AI — FastAPI, Python, OpenAI, Gemini, Supabase *Jun. 2026*

- **AI Test Generation** Built a full-stack platform generating unit tests from Python source code (AST analysis) and user stories using GPT-4.1-mini with Gemini fallback.
- **Execution Engine** Integrated pytest and pytest-cov via FastAPI to execute tests in isolated environments, returning pass/fail status and line-level coverage reports.
- **Analytics Pipeline** Persisted generated tests, execution logs, and coverage reports in Supabase (PostgreSQL), powering a historical analytics dashboard.

StealthPay — MPC, Web3, Node.js, EIP-5564, secp256k1 *Mar. 2026*

- **Privacy-Preserving Payments** Built an Ethereum dApp implementing EIP-5564 stealth addresses with a Next.js frontend and Node.js blockchain scanner.
- **Protocol Engineering** Implemented ECDH-based key derivation, stealth meta-address registry support, and ephemeral key handling compliant with EIP-5564/EIP-6538.
- **Recognition** Recognized by BitGo engineers at EthMumbai 2026 for the project's stealth address protocol implementation.

LinkStore — Next.js, TypeScript, Supabase, Cheerio *Jan. 2026*

- **Link Management** Developed a full-stack bookmark manager with authenticated CRUD operations, category organization, and Row-Level Security using Supabase.
- **Metadata Engine** Built a resilient OG metadata scraper using Cheerio with Open Graph, Twitter Card, and HTML fallbacks for rich link previews.
- **Performance & Security** Implemented multi-level sorting, pinning, XSS sanitization with DOMPurify, and strict Content Security Policy headers for secure rendering.

Education

M.Sc. in Computer Science (Blockchain Technology) *Jun. 2025 – Aug. 2027*
MIT World Peace University, Pune, Maharashtra

B.Sc. in Computer Science — CGPA: 7.00 *Jun. 2021 – Apr. 2024*
K.M.C. College of Arts, Science and Commerce, Khopoli

Technical Skills

Languages: TypeScript, JavaScript, Python, C++

Frontend: React, Next.js, Tailwind CSS, shadcn/ui

Backend: Node.js, FastAPI, Express, REST APIs, Microservices, JWT Authentication

Databases: PostgreSQL, Supabase, MongoDB, Prisma ORM, pgvector

AI & Agents: RAG, LLM Integration, n8n, MCP

Blockchain & Cryptography: Ethereum, Solidity, ethers.js, zkSync Era, Bitcoin, Stealth Addresses, Wallet Integration, zkSNARKs, ECDH, Dilithium, Post-Quantum Cryptography

Tools & Platforms: Linux, Git, Docker, GitHub Actions, YAML, Vercel, Nosana CLI, Hugging Face, Postman